



Blinkit Data Analytics Capstone Project

Insights from SQL, Power BI, and Python

Mentorship Program: WsCube Tech – Data Analytics

By- Shrotashrini Bhattacharjya

PROBLEM STATEMENT

Blinkit, a leading quick-commerce grocery delivery platform, seeks to optimize its end-to-end business operations by analyzing customer purchasing behavior, delivery performance, inventory management, marketing effectiveness, and customer satisfaction metrics.

The goal is to identify growth levers, improve operational efficiency, personalize customer engagement, and build predictive models for delivery time estimation, customer segmentation, and sales forecasting to drive profitable and scalable growth.

This capstone simulates a real-world, end-to-end analytics project, combining advanced SQL querying, interactive dashboards, and machine learning for predictive modeling.

TABLE OF CONTENTS

1. PROJECT OVERVIEW
2. PROJECT OBJECTIVES
3. ALL SQL QUESTIONS (Query → Output → Key Insights → Recommendations)
4. Multi-page Power BI Dashboard Screenshots (Key Insights → Recommendations)
5. Machine Learning Overview

PROJECT OBJECTIVES

❑SQL Insights Extraction

Use advanced SQL to extract business-critical insights from Global Fashion Retail's operations, enabling leadership to make informed strategic decisions.

❑Interactive Dashboard

Build a multi-page Power BI dashboard to visualize sales trends, product performance, promotions, and customer behavior for executive-level analytics.

❑Machine Learning Applications

Develop ML models to predict returns and forecast demand, optimizing operations and inventory management.

Identify the top 10 customers by total order value.

```
SELECT
    customer_id, customer_name, total_orders, avg_order_value,
    round(total_orders * avg_order_value,2) AS total_order_value
FROM
    customers
ORDER BY total_order_value DESC
LIMIT 10;
```

Count the number of customers in each segment (Premium, Regular, Inactive, New)

```
SELECT DISTINCT
    customer_segment, COUNT(*) AS no_of_customers
FROM
    customers
GROUP BY customer_segment;
```

Find customers with an average order value above 500 and more than 10 orders

```
SELECT
    customer_id, customer_name, avg_order_value
FROM
    customers
WHERE
    avg_order_value > 500
    AND total_orders > 10;
```

Key Insights

1. Revenue Concentration

A small group of customers contributes a disproportionately high share of total revenue. These are likely your high-value / premium customers.

2. Customer Segment Distribution

- The count per segment helps reveal:
- Whether your business is growing (more New customers)
- Retention strength (Premium/Regular size)
- Churn risk (Inactive customers)

3. High-Value Engaged Customers

Customers with Avg order value > 500 and more than 10 orders are the most loyal and profitable segment.

Recommendations

1. Strengthen High-Value Customer Retention

Offer:

- Loyalty programs
- Exclusive discounts
- Early access perks

Goal: **Prevent churn of top contributors**

2. Upsell & Convert Regular Customers

Target Regular → Premium:

- Personalized offers
- Bundling strategies
- Spend-based rewards

3. Re-engage Inactive Customers

Run:

- Win-back campaigns
- Email/SMS reminders
- Special comeback offers

Goal : **Identify *why* they churned (price, experience, delivery, etc.)**

List orders that were delivered late along with reasons for delay.

```
SELECT
    order_id, reasons_if_delayed
FROM
    delivery_performance
WHERE
    delivery_status != 'On Time'
    OR delivery_status IS NULL;
```

Calculate the average delivery time (in minutes) per delivery partner

```
SELECT
    delivery_partner_id,
    AVG(delivery_time_minutes) AS avg_delivery_in_minutes
FROM
    delivery_performance
GROUP BY delivery_partner_id;
```

Find the top 5 stores by total order revenue

```
SELECT
    store_id,
    SUM(order_total) AS total_order_revenue
FROM
    orders
GROUP BY store_id
LIMIT 5;
```

Key Insights

1. Late Deliveries & Reasons

- We are capturing all orders where delivery was **not “On Time” or missing status**.
- This helps identify:
 - **Operational bottlenecks** (e.g., traffic, preparation delay)
 - **Data quality issues** (NULL statuses).

2. Delivery Partner Performance (Avg Delivery Time)

- Average delivery time per partner reveals:
 - **Top performers (fast delivery)**
 - **Underperformers (slow delivery)**
- High variation between partners = **inconsistent service quality**.

3. Top 5 stores by revenue

These stores likely:

- Have **high demand**
- Are in **prime locations**
- Offer **better pricing/quality**

Recommendations

1. Fix Delivery Delays (Root Cause Analysis)

- Break down delay reasons:
 - Store preparation delay
 - Delivery partner delay
 - External factors (traffic/weather)
- Prioritize **top recurring issues**

2. Support High-Revenue Stores

- Top stores are critical:
 - Ensure they have **adequate delivery capacity**
 - Monitor their **prep time + order load**
- Avoid delays caused by **overload**

3. Optimize Delivery Partner Allocation

- **For slow partners:**
 - Provide training or route optimization
 - Reduce order assignment temporarily
- **For fast partners:**
 - Incentivize with bonuses
 - Assign high-priority orders

Identify products that have had damaged stock more than 5 times in total.

```
with all_inventory as ( select product_id, damaged_stock from inventory union all select product_id, damaged_stock from inventorynew)
SELECT product_id,
       SUM(damaged_stock) AS total_damage_stock
FROM all_inventory
GROUP BY product_id
HAVING SUM(damaged_stock) >5;
```

Key Insights

1. Recurring Product-Level Damage

- Products appearing in this list have **systematic damage issues**, not one-off incidents.
- Indicates problems like:
 - Fragile product design
 - Poor packaging
 - Repeated handling/storage issues

2. Cost & Profit Impact

- Frequent damaged stock leads to:
 - **Direct revenue loss**
 - Increased **replacement/refund costs**
 - Higher **inventory shrinkage**

3. Potential Process Gaps

High damage counts may indicate:

- Poor warehouse handling
- Inefficient logistics
- Supplier quality issues

Recommendations

1. Root Cause Analysis (Priority Action)

- For each high-damage product:
 - Check **where damage occurs**:
 - Supplier
 - Warehouse
 - Transit
- Tag damage by stage if not already tracked.

2. Improve Packaging & Product Design

- For fragile items:
 - Reinforce packaging
 - Add protective layers
- Work with suppliers to:
 - Improve material quality
 - Reduce defect rates.

3. Supplier Performance Monitoring

- Identify suppliers linked to high-damage products
- Introduce:
 - Quality checks
 - Penalties for defective shipments

Calculate the total quantity ordered for each product

```
SELECT product_id,  
       SUM(quantity) AS total_quantity  
FROM order_items  
GROUP BY product_id;
```

Find products that often fall below the minimum stock level (compare current stock with min stock).

```
SELECT product_id, SUM(damaged_stock) AS total_damage_stock FROM all_inventory  
GROUP BY product_id  
HAVING SUM(damaged_stock) >5;  
SELECT product_id, SUM(quantity) AS total_quantity  
FROM order_items  
GROUP BY product_id;  
WITH combined_inventory AS (SELECT product_id, date, stock_received, damaged_stock FROM inventory  
                             UNION ALL  
                             SELECT product_id, date, stock_received, damaged_stock FROM inventorynew  
                             )  
, stock_calc AS (  
    SELECT product_id, date, (stock_received - damaged_stock) AS effective_stock FROM combined_inventory  
)  
SELECT p.product_id, p.product_name, COUNT(*) AS times_below_min FROM stock_calc as sc  
JOIN products as p  
    ON sc.product_id = p.product_id  
WHERE sc.effective_stock < p.min_stock_level  
GROUP BY p.product_id, p.product_name  
HAVING COUNT(*) > 1  
ORDER BY times_below_min DESC;
```

Key Insights

1. Product Demand (Total Quantity Ordered)

- It helps identify:
 - **Fast-moving items**
 - Core revenue drivers
- Likely follows a pattern where a small number of products generate **most of the demand**.

2. Frequent Stockouts (Below Minimum Stock Level)

- It indicates:
 - **Poor inventory planning**
 - **Demand-supply mismatch**
 - Inefficient replenishment cycles
- Insight: These products face **recurring stock shortages**, not occasional dips.

Recommendations

1. Prioritize High-Demand, Low-Stock Products

- Identify overlapping products:
 - High total_quantity
 - High times_below_min
- Actions:
 - Increase safety stock
 - Shorten replenishment cycles

2. Improve Replenishment Strategy

- Automate reordering when stock reaches threshold
- Reduce supplier lead times
- Maintain buffer stock for fast-moving items

3. Stock Monitoring Dashboard

Track:

- Stockout frequency
- Product demand trends
- Inventory turnover
- Fill rate (orders fulfilled vs requested)

Calculate total revenue generated and ROAS for each marketing campaign.

```
SELECT
    campaign_name,
    ROUND(SUM(revenue_generated), 2) AS total_rev_generated,
    ROUND((SUM(revenue_generated) / SUM(spend)), 2) AS total_roas
FROM
    marketing_performance
GROUP BY campaign_name;
```

Find the campaign with the highest conversion rate (conversions/impressions)

```
SELECT campaign_id, campaign_name,
       concat((round(conversions * 1.0 / NULLIF(impressions, 0), 2)) * 100, "%") AS conversion_rate
FROM marketing_performance
ORDER BY conversion_rate DESC
LIMIT 1;
```

Key Insights

1. Revenue Contribution by Campaign

This reveals:

- Which campaigns drive the **most sales**
- Dependence on a few high-performing campaigns

2. Conversion Rate Leader

- Conversion rate = conversions / impressions
- Identifies the campaign that is:
 - Best at **turning interest into action**
- This campaign likely has:
 - Strong targeting
 - Effective creatives
 - Good audience-product fit

Recommendations

1. Scale High-ROAS Campaigns

Increase budget for:

- Campaigns with **strong ROAS**
- These are your **most profitable growth levers**

2. Leverage High Conversion Campaigns

For campaigns with top conversion rates:

- Increase impressions (budget scaling)
- Replicate strategy across other campaigns

3. Campaign Segmentation Strategy

Classify campaigns into:

- High ROAS + High Revenue → **Scale aggressively**
- High ROAS + Low Revenue → **Increase budget**
- Low ROAS + High Revenue → **Optimize**
- Low ROAS + Low Revenue → **Pause or rethink**

List all campaigns targeted at Premium customers and their performance metrics.

```
SELECT campaign_id, campaign_name, date, channel,  
    -- Base Metrics  
    impressions, clicks, conversions, spend, revenue_generated,  
    -- Performance Metrics  
    CONCAT(ROUND((clicks * 1.0 / NULLIF(impressions, 0)) * 100, 2), '%') AS CTR,  
    CONCAT(ROUND((conversions * 1.0 / NULLIF(impressions, 0)) * 100, 2), '%') AS conversion_rate,  
    ROUND(revenue_generated * 1.0 / NULLIF(spend, 0), 2) AS ROAS  
FROM marketing_performance  
WHERE target_audience = 'Premium'  
ORDER BY  
    (conversions * 1.0 / NULLIF(impressions, 0)) DESC;
```

Key Insights

1. Premium Audience = High-Value Segment

- These campaigns target customers with:
 - Higher spending capacity
 - Stronger lifetime value (LTV)
- Performance here directly impacts **profitability**, not just volume.

2. Channel Effectiveness

- Inclusion of channel helps identify:
 - Which platforms work best for premium users (e.g., search, social, email)
- Premium audiences often respond better to:
 - Personalized, high-quality channels.

3. Conversion Rate–Driven Ranking

- Campaigns are ordered by **conversion rate**, highlighting:
 - Most effective campaigns at **turning impressions into purchases**
- High conversion rate suggests:
 - Strong messaging
 - Precise targeting
 - High intent audience

Recommendations

1. Scale Top Premium Campaigns

- Increase budget for campaigns with:
 - High conversion rate
 - Strong ROAS
- These deliver **maximum value per impression**.

2. Focus on High-ROAS Campaigns

Prioritize campaigns that generate **more revenue per dollar spent**

3. Personalization Strategy

- Premium customers respond well to:
 - Exclusive offers
 - Personalized messaging
 - Loyalty benefits
- Tailor campaigns accordingly

**Count feedback entries by sentiment
(Positive, Neutral, Negative)**

```
SELECT
    sentiment, COUNT(*) AS feedback_counts
FROM
    customer_feedback
GROUP BY sentiment;
```

**List customers who gave negative feedback and their
corresponding orders.**

```
SELECT
    c.customer_id, c.customer_name,
    COUNT(cf.order_id) AS negative_orders_count
FROM customers c
JOIN customer_feedback cf
    ON c.customer_id = cf.customer_id
WHERE cf.sentiment = 'Negative'
GROUP BY c.customer_id, c.customer_name;
```

Key Insights

1. Overall Customer Sentiment Health

- The sentiment distribution (**Positive, Neutral, Negative**) provides a clear view of **customer satisfaction**.
- Key patterns:
 - High **Positive** → strong brand experience
 - High **Neutral** → average experience, improvement opportunity
 - High **Negative** → service/product issues

2. Customer Experience Breakdown

- Two possible patterns:
 - **High-order customers + negative feedback** → declining experience → high churn risk
 - **Low-order customers + negative feedback** → poor first experience

Negative feedback affects both **retention and acquisition**.

Recommendations

1. Target High-Risk Customers Immediately

- Identify customers with:
 - Multiple negative feedback entries
 - High order volume
- Actions:
 - Personalized outreach
 - Issue resolution
 - Compensation or retention offers

2. Categorize and Fix Root Causes

- Break down negative feedback into:
 - Delivery issues
 - Product defects
 - Service complaints
- Focus on **top recurring problem areas**

Customer Feedback & Sentiment Analysis

Total Feedback
5000

Negative Feedback
1642

Avg Rating
3.34

feedback_id
All

sentiment
All

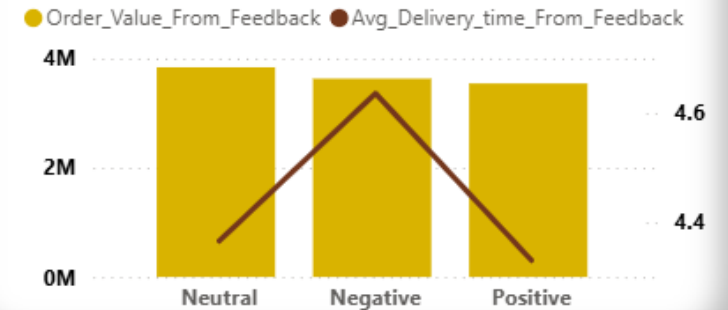
Top Customer Feedback Themes



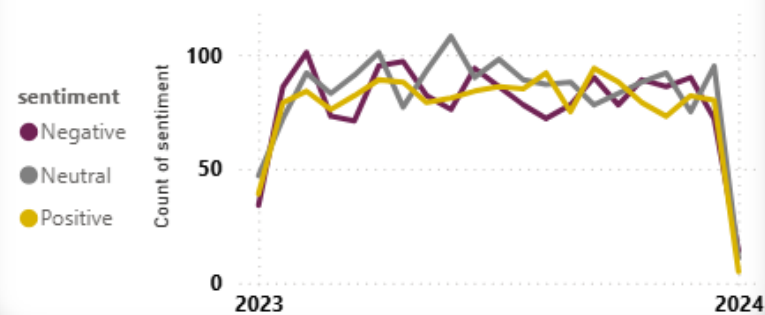
Customer Feedback & Rating by Category



Impact of Delivery Performance on Customer Sentiment



Customer Sentiment Trend Over Time



Negative Feedback by Root Cause



Key Insights

1. Customer Distribution is Balanced Across Segments

- Segments (Regular, Premium, New, Inactive) are **almost evenly distributed (~24–26%)**
- Growth must come from **conversion (New → Premium)** rather than acquisition alone.

2. New Customers Drive the Highest Order Volume

- **New segment shows highest total orders**
- Strong acquisition, but **weak conversion into loyal/high-value customers.**

3. Geographic Concentration

- Certain areas (e.g., Jalna, Bathinda, Deoghar) show higher customer counts.
- Demand is **regionally concentrated**, not evenly distributed.

Recommendations

1. Convert New → Premium (Top Priority)

- Introduce:
 - Loyalty programs
 - First 3-order incentives
 - Subscription models

Goal: Increase **customer lifetime value (LTV)**

2. Upsell Regular Customers

- Strategies:
 - Personalized recommendations
 - Bundle offers
 - Spend-based rewards

Goal: Move Regular → Premium

3. Regional Expansion Strategy

Focus on:

- High-performing areas → scale further
- Low-performing areas → targeted campaigns

Order and Delivery Performance

69.40%

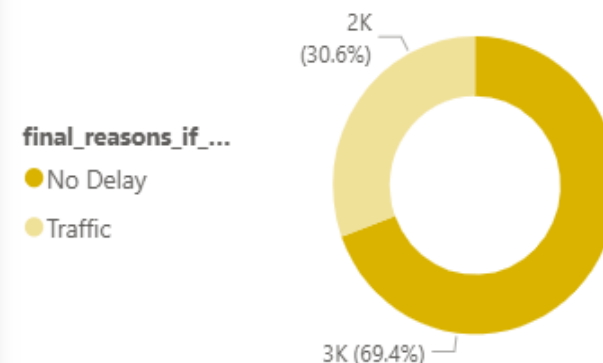
On_time_Pct

30.60%

Delayed_Pct

area	Avg_Delivery_Time
Adoni	5.22
Agartala	4.89
Agra	7.19
Ahmedabad	2.25
Ahmednagar	3.33
Aizawl	5.08
Ajmer	4.08
Akola	4.33
Alappuzha	6.27
Aligarh	5.22
Allahabad	4.13
Alwar	0.40
Amaravati	1.14
Ambala	3.55
Total	4.44

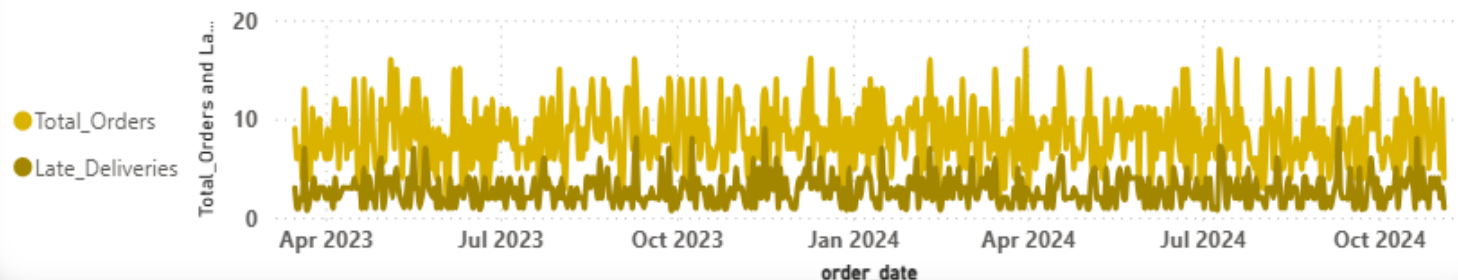
Total_Orders by final_reasons_if_delayed



Delayed_orders by final_reasons_if_delayed



Total_Orders and Late_Deliveries by order_date



Key Insights

1. On-Time Performance is Moderate (Needs Improvement)

- **On-time deliveries: 69.4%**
- **Delayed deliveries: 30.6%**
- A ~30% delay rate is **too high for a last-minute delivery model**. This directly impacts customer satisfaction and retention.

2. Traffic is the Dominant Cause of Delays

- Majority of delays are attributed to **Traffic**
- Other causes are minimal or not captured.

3. Average Delivery Time Varies Significantly by Area

- Some areas: Very fast (e.g., ~0.4–2 mins)
- Others: Much slower (5–7+ mins)
- Delivery performance is **location-dependent**, likely due to:

Traffic density, Distance ,Partner availability

Recommendations

1. Reduce Delay Rate (Top Priority)

- Target: Bring delays from **30% → <15%**
- Actions:
 - Improve routing algorithms
 - Increase delivery partner availability during peak hours.

2. Optimize for Traffic Conditions

- Since traffic is the main issue:
 - Use **real-time traffic-based routing**
 - Predict peak congestion hours
- Consider:
 - Micro-warehousing in dense areas.

3. Improve Delay Categorization

- “Traffic” alone is too broad
- Add granular reasons:
 - Partner delay
 - Store prep delay
 - Weather
 - System issues
- Better data = better decisions

Marketing Campaign Performance

Total Rev
32.19M

Total Spend
16.32M

Total Clicks
3M

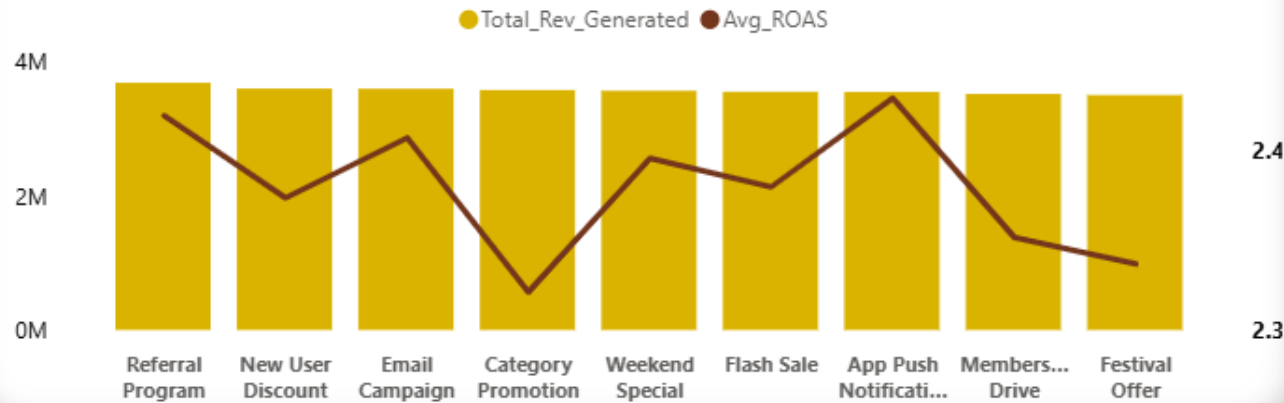
Conversions
298K

campaign_id
All

campaign_name
All

channel
All

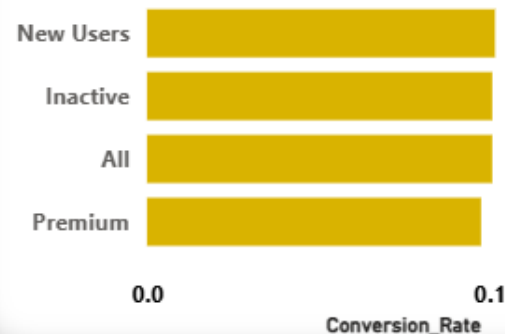
Campaign Revenue & ROAS Analysis



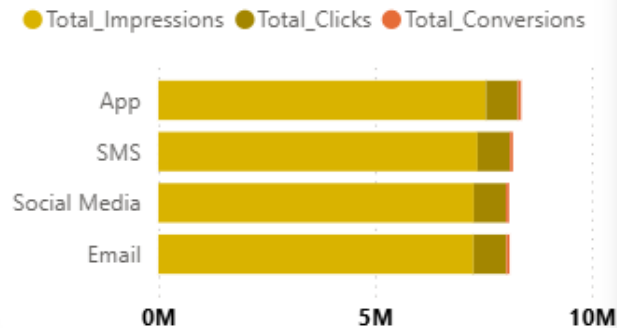
Campaign Performance Summary

campaign_name	Total_Impressions	Total_Clicks	Total_Conversions	Total_Spend
Category Promotion	3323234	329690	33568	1,850,583.00
New User Discount	3270347	332400	33964	1,833,454.81
Flash Sale	3264022	332882	33211	1,831,687.82
Referral Program	3276426	329302	33490	1,818,025.51
Email Campaign	3292685	338600	32436	1,810,729.67
Weekend Special	3290521	323668	33271	1,799,611.29
Festival Offer	3306651	325928	32765	1,796,687.14
Membership Drive	3333717	335842	32550	1,790,069.80
App Push Notification	3130007	325833	32783	1,788,989.20

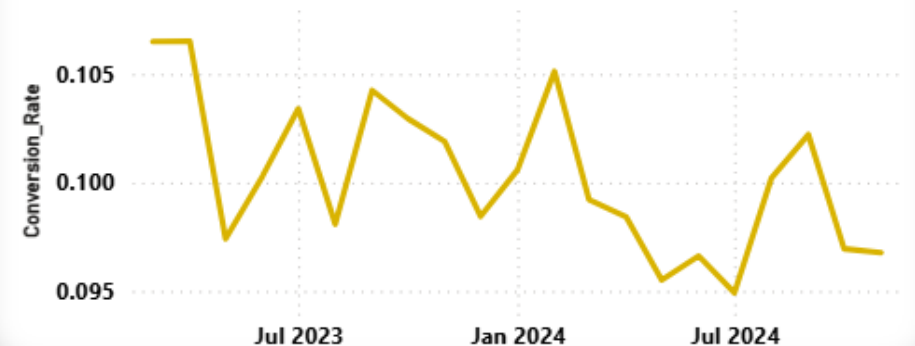
Audience Segment Performance Analysis



Channel-wise Marketing Performance



Conversion Rate Trend Over Time



Key Insights

1. Strong Overall Revenue but Efficiency Needs Attention

- **Total Revenue:** 32.19M
- **Total Spend:** 16.32M
- → Approx **ROAS** $\approx 2x$
- You are profitable, but **not highly optimized**—there's room to push ROAS higher.

2. Conversions vs Clicks Gap

- **Clicks:** 3M
- **Conversions:** 298K (~10% conversion rate)
- Insight:
- Decent conversion rate, but:
- ~90% of traffic is **not converting** → opportunity for funnel optimization

3. Campaign Performance is Uneven

- Campaigns like:
 - **App Push Notifications / Referral Program** → higher ROAS
- Campaigns like:
 - **Category Promotion** → lower ROAS
- Insight:

Not all campaigns are equally efficient → budget allocation can be improved

Recommendations

1. Segment-Based Targeting

Customize campaigns:

- New users → onboarding offers
- Premium users → exclusive deals
- Inactive users → reactivation campaigns

2. Fix Underperforming Campaigns

For low ROAS campaigns:

- Improve creatives
- Refine audience targeting
- Test new messaging

3. Reallocate Budget Based on ROAS

- Increase spend on:
 - High-ROAS campaigns (e.g., App Push, Referral)
- Reduce spend on:
 - Low-ROAS campaigns (e.g., Category Promotion)

4. Optimize Conversion Funnel

- Improve:
 - Landing pages
 - Checkout experience
 - Call-to-actions

Goal: Convert more of the **existing 3M clicks**



Product & Inventory Management

Product Counts
Below Min Stock

0

product_id

All

product_name

All

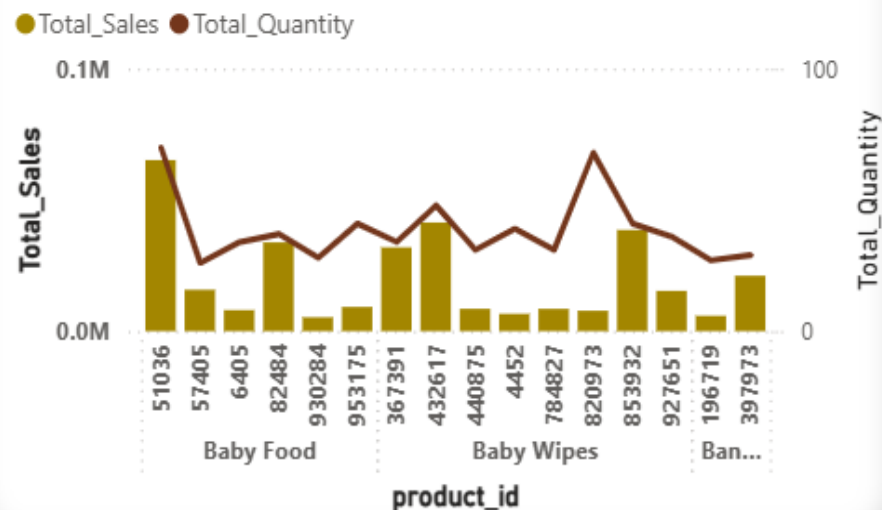
category

All

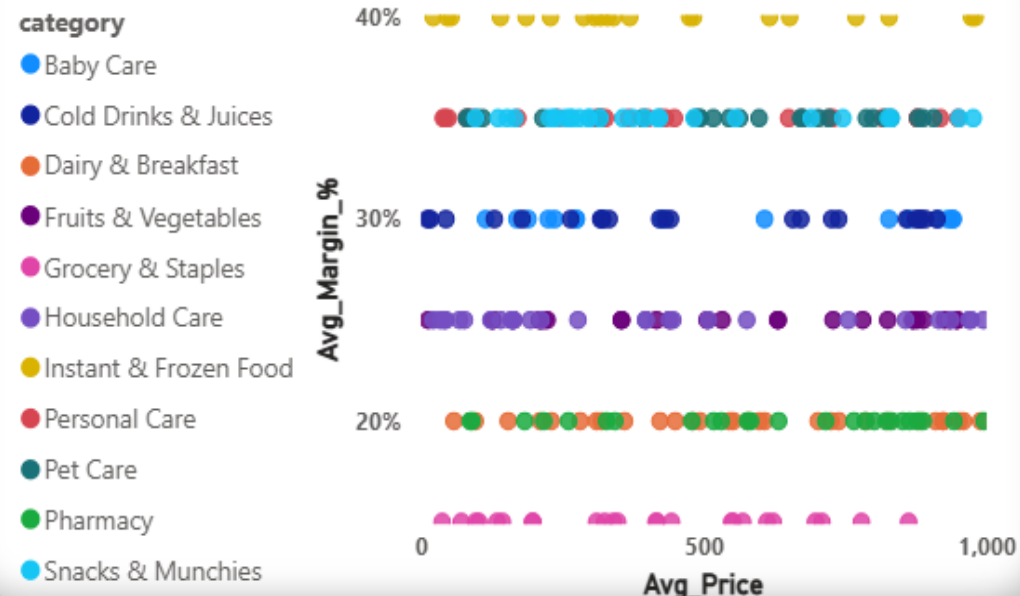
brand

All

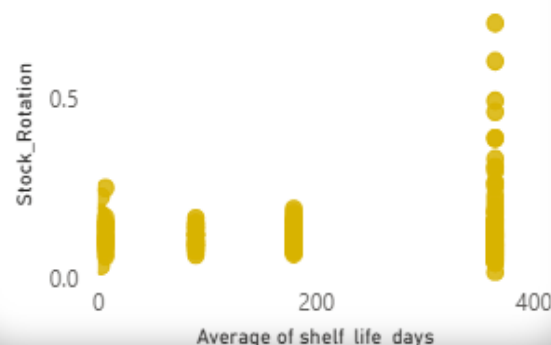
Total_Sales and Total_Quantity by product_name and product_id



Avg_Price and Avg_Margin_% by product_display and category

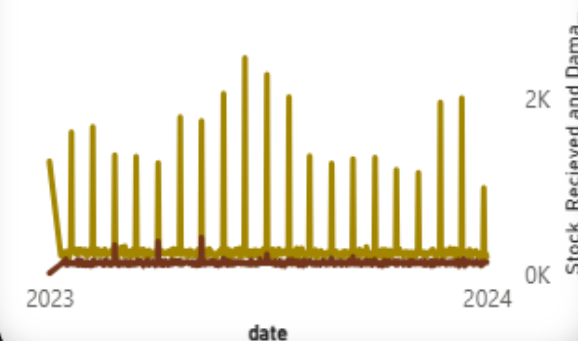


Average of shelf_life_days and Stock_Rotation by product_display



product_id	Effective_Stock	Min_Stock_Level	Below_Min_S
107548	311	27	Sufficient
112390	312	23	Sufficient
11422	299	20	Sufficient
114414	439	26	Sufficient
118820	336	12	Sufficient
120385	523	28	Sufficient
120532	652	29	Sufficient
120532	652	29	Sufficient
120532	652	29	Sufficient
Total	94928	5464	Sufficient

Stock_Rcieved and Damaged_Stock by date



Key Insights

1. Wide Price vs Margin Variation

- Products span:
 - ₹0–₹1000 price range
 - ~0%–40% margin
- Insight:
- No consistent pricing strategy
- Some high-priced products have **low margins**.

2. Damaged Stock Exists but Not Dominant

- Damaged stock is present but relatively low compared to stock received
- Insight:
- Damage control is decent
- Still an opportunity to reduce waste further

3. No Immediate Stock Risk (Good Sign)

- **Products below minimum stock: 0**

Insight:

Inventory is currently **well-managed and sufficiently stocked** across products.

But:

This could also indicate **overstocking**, not just efficiency.

Recommendations

1. Optimize Inventory Levels

- Reduce excess stock for:
 - Slow-moving products
- Align stock with:
 - Demand patterns

Goal: Free up working capital

2. Identify & Fix Slow-Moving Products

Take action:

- Discounts / clearance sales
- Bundle offers
- Reduce future procurement

3. Improve Pricing Strategy

For:

- High-demand products → increase margin slightly
- Low-demand products → adjust pricing or promote

4. Balance Margin vs Volume

Focus on:

- Products that have:
 - Good sales volume
 - Good margins

These should be the **core products**



Customer Feedback & Sentiment Analysis

Total Feedback
5000

Negative Feedback
1642

Avg Rating
3.34

feedback_id
All

sentiment
All

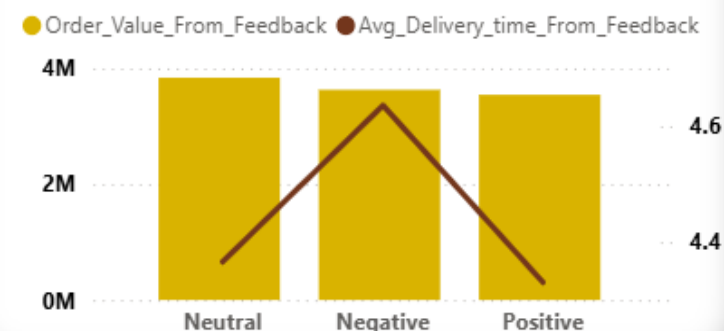
Top Customer Feedback Themes



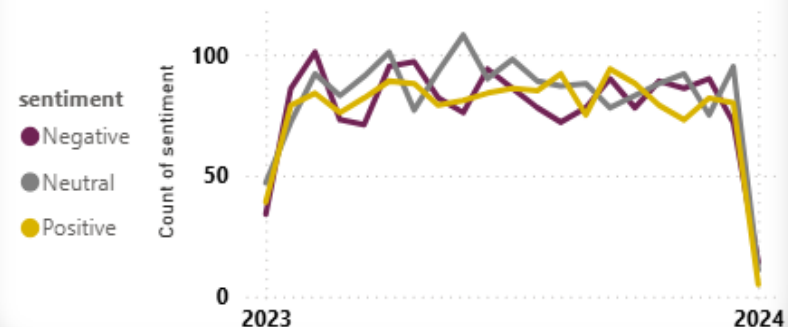
Customer Feedback & Rating by Category



Impact of Delivery Performance on Customer Sentiment



Customer Sentiment Trend Over Time



Negative Feedback by Root Cause



Key Insights

1. High Negative Feedback Share (~33%)

- **Total Feedback:** 5000
- **Negative Feedback:** 1642
- → ~1 in 3 customers is unhappy
- Insight:
This is **significantly high** and signals underlying experience issues.

2. Average Rating is Low (3.34 / 5)

- Indicates **mediocre customer satisfaction**
- Not bad enough to churn immediately, but **not strong enough for loyalty**

3. Top Feedback Themes Highlight Core Issues

- Most common keywords:
 - *Product*
 - *Delivery*
 - *Experience*
 - *Service*

Insight:

Problems are **operational, not cosmetic.**

Core business functions are driving dissatisfaction

Recommendations

1. Fix Delivery Experience (Top Priority)

- Reduce delivery delays
- Improve real-time tracking
- Direct impact on sentiment & retention

2. Improve Product Quality Control

- Address:
 - Damaged items
 - Incorrect products
- Strengthen:
 - Supplier quality checks
 - Packaging standards

3. Resolve “No Delivery Data” Issue

- Fix:
 - Missing tracking data
 - Logging/ETL issues
- Ensure every order has:
 - Delivery status
 - Delay reason



Financial & Operational KPIs

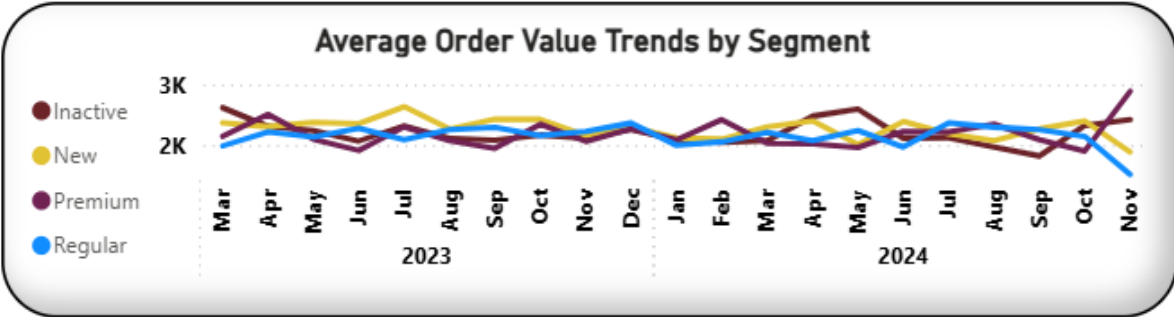
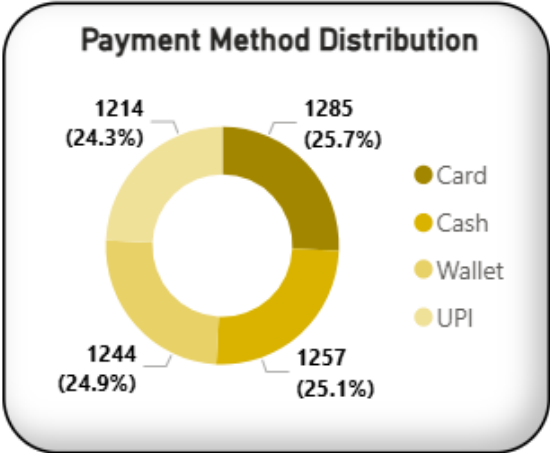
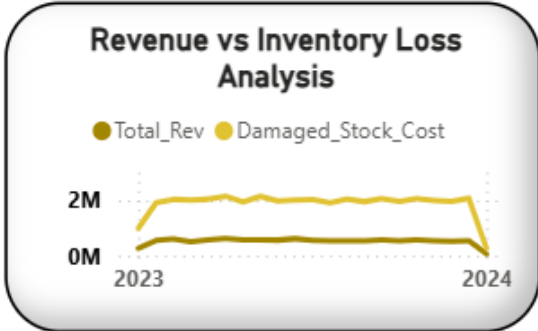
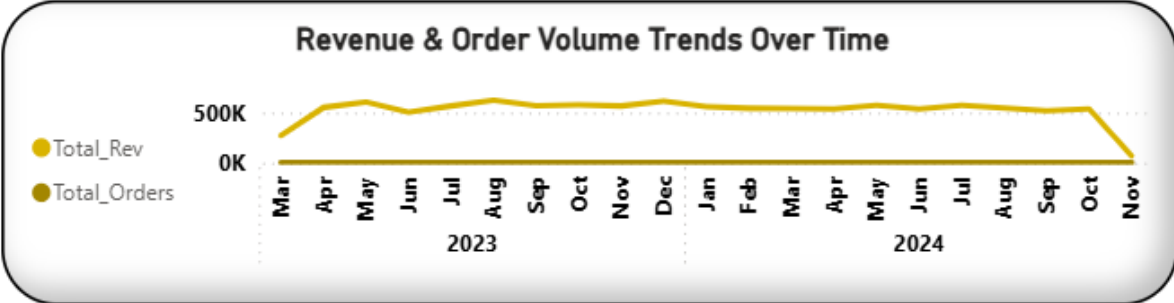
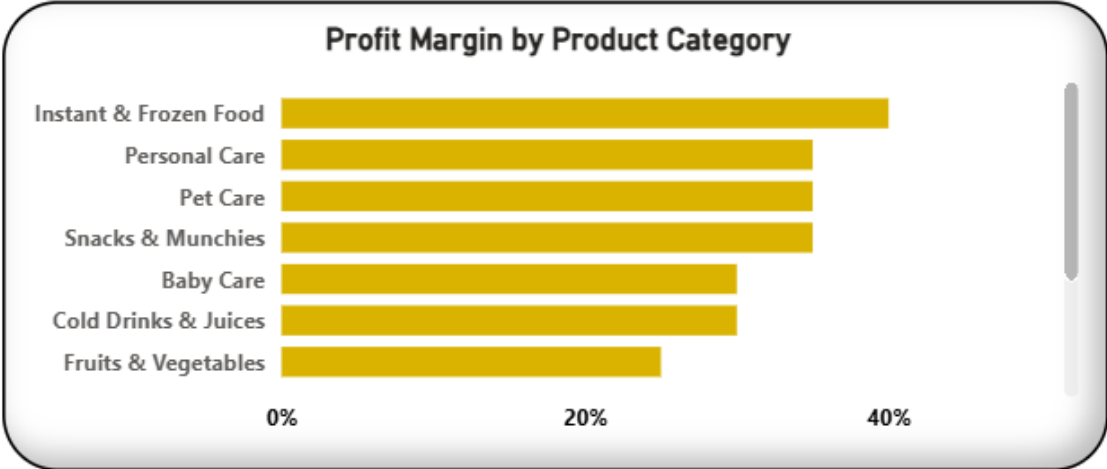
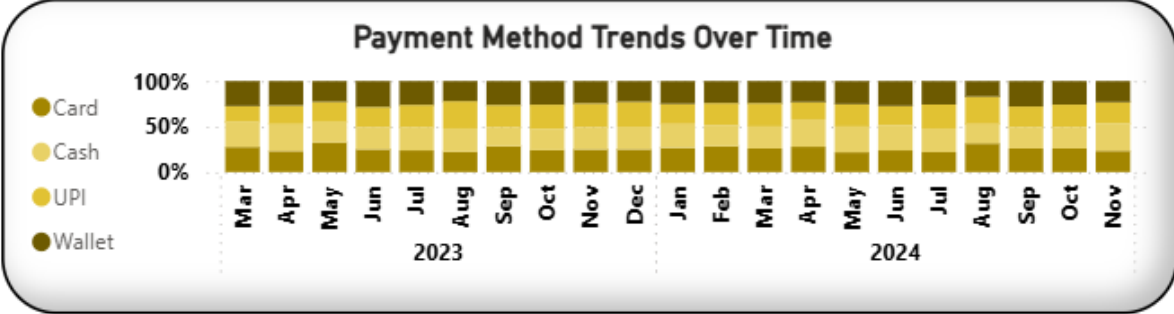
Total Damaged Units
81K

Damage Rate
46.16%

payment_method
All

category
All

Date
All



Damaged stock cost is cumulative over time, while revenue reflects only transactional sales.

Key Insights

1. Extremely High Damage Rate (Major Concern)

- Damaged Units: 81K
- Damage Rate: 46.16%
- Nearly half of your inventory is getting damaged — this is a critical operational issue, not a minor inefficiency.

2. Inventory Loss is Eating Into Revenue

- Revenue vs damaged stock trend shows:
 - Revenue is steady
 - But damaged stock cost is accumulating over time
 - Revenue is being earned but losing profitability due to inventory damage.

3. Profit Margins Vary by Category

- High margins:
 - Instant & Frozen Food
 - Personal Care
- Low margins:
 - Fruits & Vegetables
- Insight:
- Some categories are **profitable**
- Others are **volume-driven but low-margin**

Recommendations

1. URGENT: Reduce Damage Rate (Top Priority)

- 46% is extremely high → needs immediate action
- Fix:
 - Storage conditions
 - Packaging quality
 - Handling processes
 - Supplier quality

2. Track Damage by Category & Supplier

- Identify:
 - Which products are most damaged
- Take action:
 - Replace suppliers
 - Improve packaging for sensitive items

3. Protect High-Margin Categories

- Focus on:
 - Instant & Frozen Food
 - Personal Care

Ensure:

- There is low damage
- There is high availability

Machine Learning Overview

Descriptive Statistics:

- **Count-**

Number of valid (non-missing) values.

- **Interpretation:**

- If count < total rows → missing data exists
- Example: delivery_time_minutes = 3437 vs 5000
Missing values were present earlier.

- **Mean (Average)-**

Customers typically spend around ₹2200 per order.

- **Median (50%)-**

- order_total mean $\approx$ median → fairly balanced.

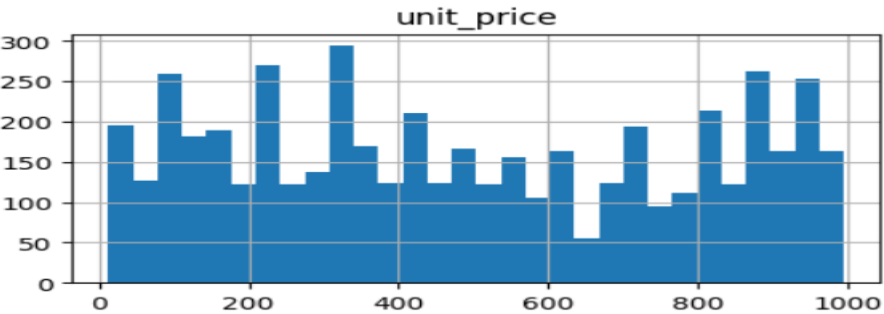
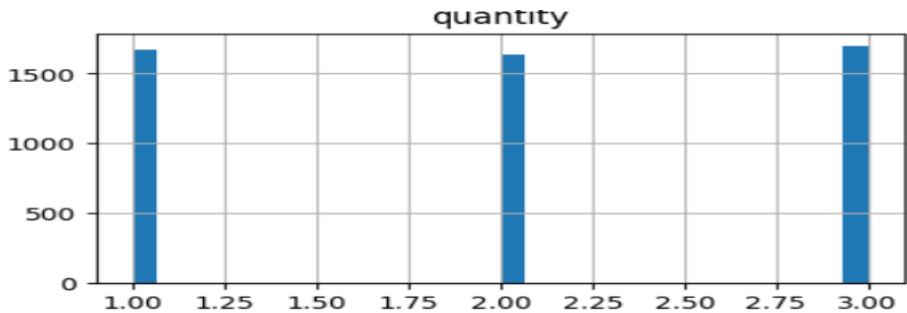
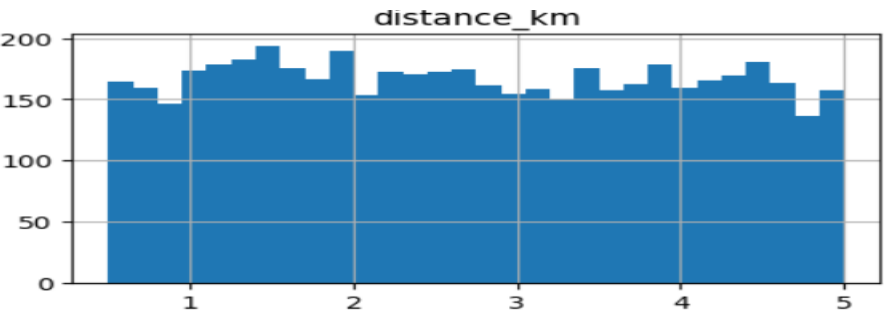
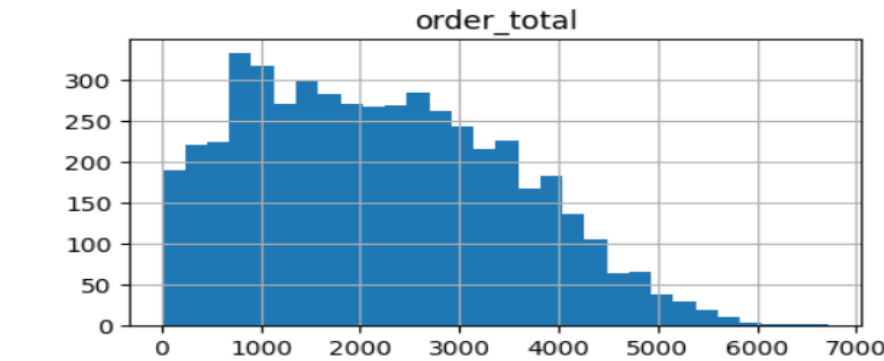
- **Standard Deviation (std)-**

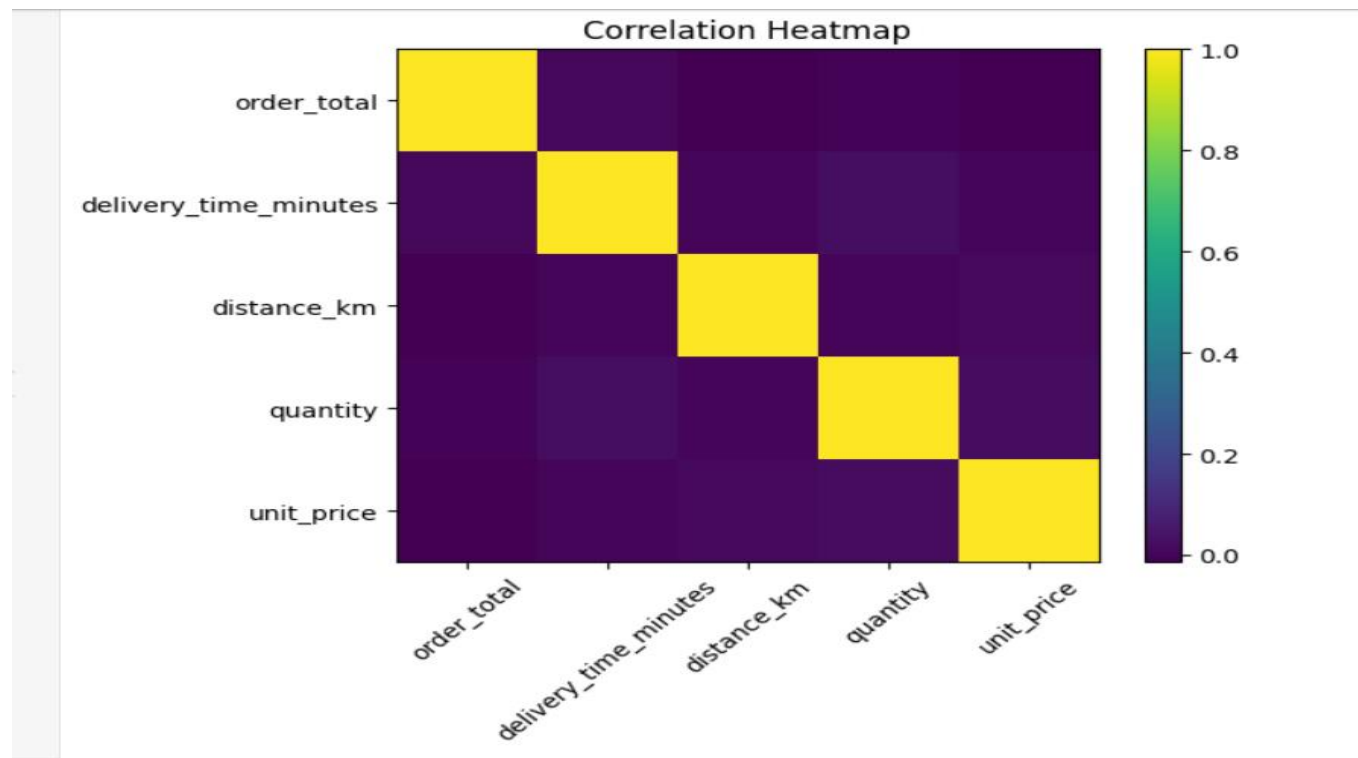
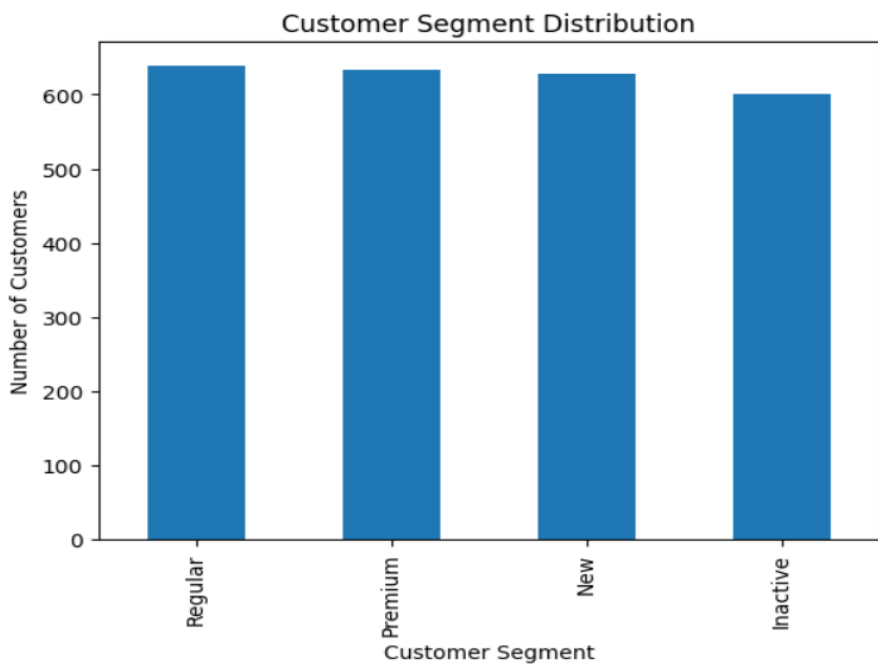
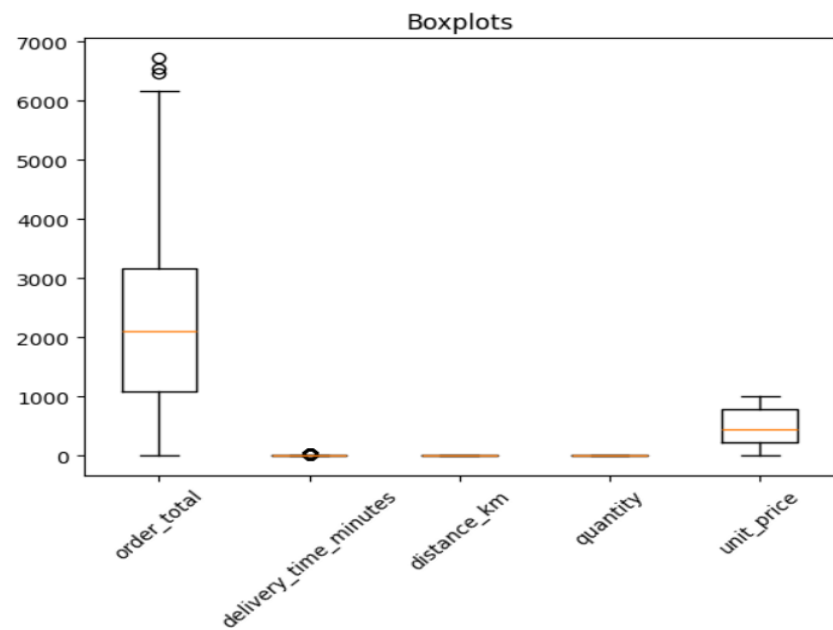
Interpretation:

- High std → values vary a lot
- Low std → values

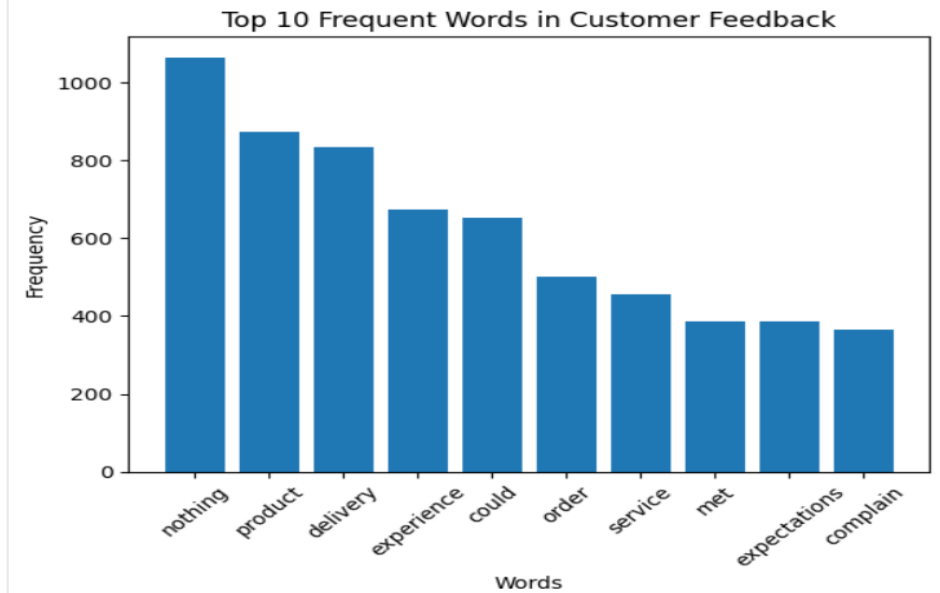
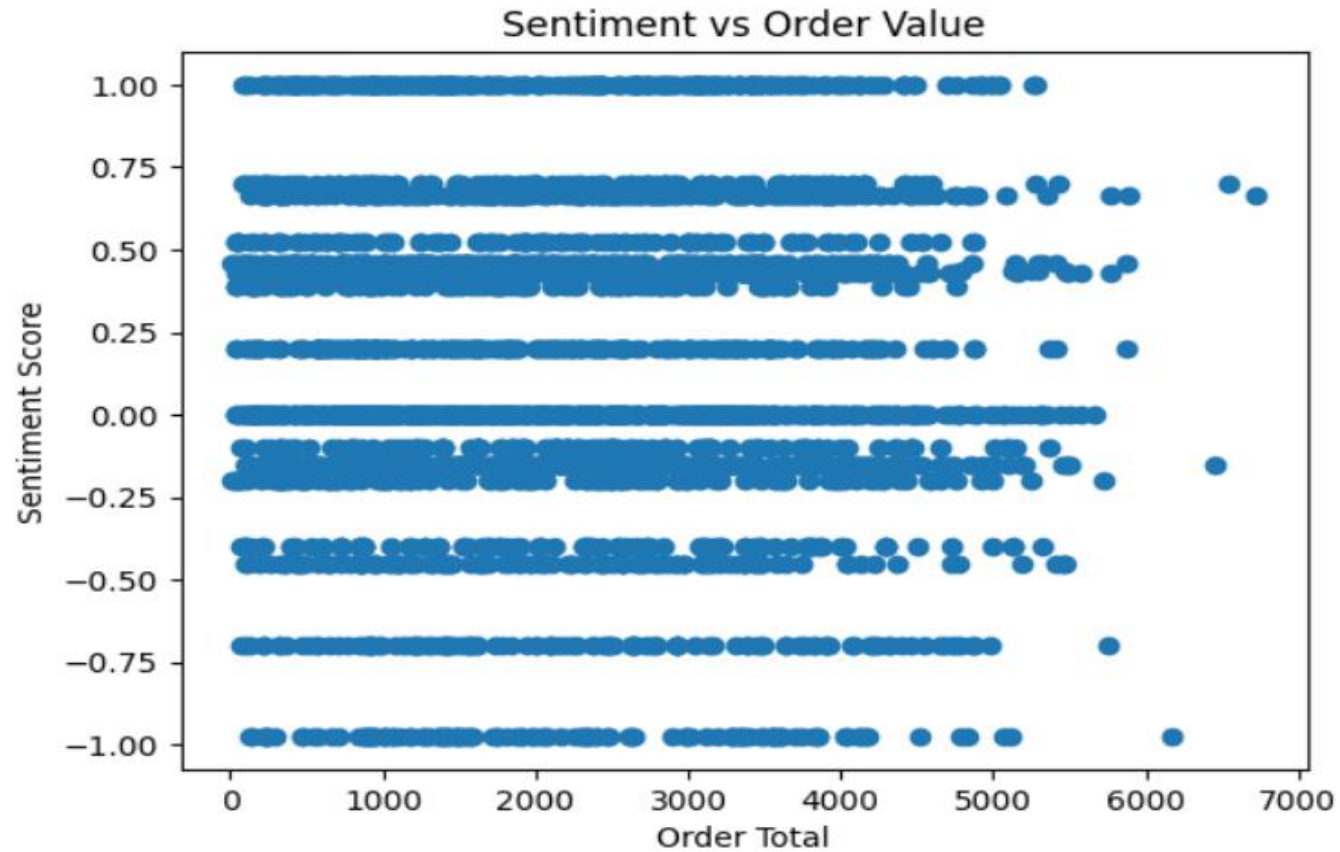
Visualizations

Data Distributions





Sentiment Analysis and text processing



Delivery Time Prediction Model

1. Accuracy Metrics:

- *The model shows relatively weak predictive performance, as the average prediction error (MAE) exceeds 3 minutes.*
- The higher RMSE compared to MAE indicates that the model occasionally makes large prediction errors. In a delivery context, such errors can lead to poor customer experience, as significant delays or early predictions may impact service reliability.

2. Feature Importance:

The regression results indicate that promised delivery time is the most significant predictor of actual delivery time, with a near one-to-one relationship. In contrast, distance has a negligible effect, suggesting that delivery operations are optimized for short distances or that its impact is already captured within the promised time. Additionally, delivery partner ID shows almost no influence, likely due to improper encoding of this categorical variable. Overall, the model relies heavily on promised time, limiting its ability to uncover deeper operational insights.

Github : [https://github.com/shrini1996/Blinkit/blob/main/Blinkit%20%20\(1\).ipynb](https://github.com/shrini1996/Blinkit/blob/main/Blinkit%20%20(1).ipynb)

Thank You!

Feel free to connect with me:

Email- Shrotashrini11@gmail.com

Github: <https://github.com/shrini1996>

Linkedin: www.linkedin.com/in/shrini199628